SMART INDIA HACKATHON 2026

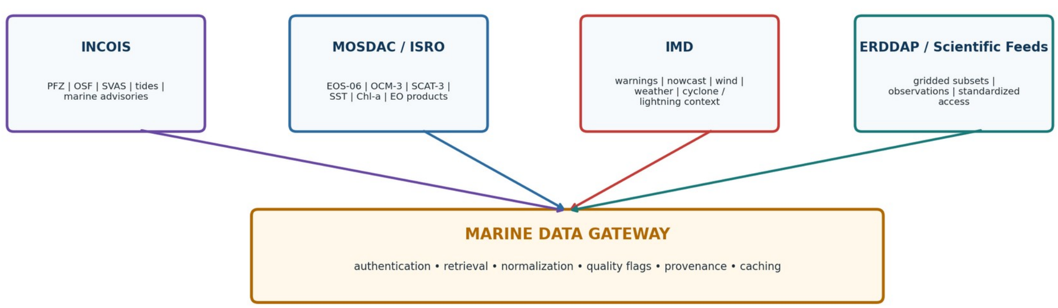
SIH 26176

ORCA

Marine Intelligence Platform

In one line: ORCA is a conversational decision-support system that connects existing marine data sources, lets specialized AI agents work on a user request, and produces a safe, explainable recommendation with a map.

Authoritative Data Ecosystem ORCA Can Orchestrate



Concept: ORCA sits above existing satellite, ocean, weather and geospatial services.

1. What is the problem statement?

ISRO is asking for an Agentic AI-powered conversational platform for marine information. The system should understand a natural-language question, decide what information is needed, retrieve data from multiple sources, reason across location and time, and return a useful recommendation.

The key point is that the solution should not be just a dashboard or a chatbot. It must connect different marine data sources and use them together.

What the user should be able to ask

- "Where is the nearest Potential Fishing Zone today?"
- "Can I safely go fishing tomorrow morning from Malpe?"
- "What are the tide, wind, wave and weather conditions near me?"
- "Which fishing areas should I avoid because of hazards or restricted waters?"

What ISRO expects the platform to demonstrate

- Natural-language understanding and multi-turn conversation.
- Autonomous planning and selection of tools/data sources.
- Collaboration between specialized agents.
- Spatial and temporal reasoning.
- Explainable recommendations with evidence, maps and alerts.
- Support for Indian regional languages.
- Safety and geofencing awareness.

2. Our proposed solution — ORCA

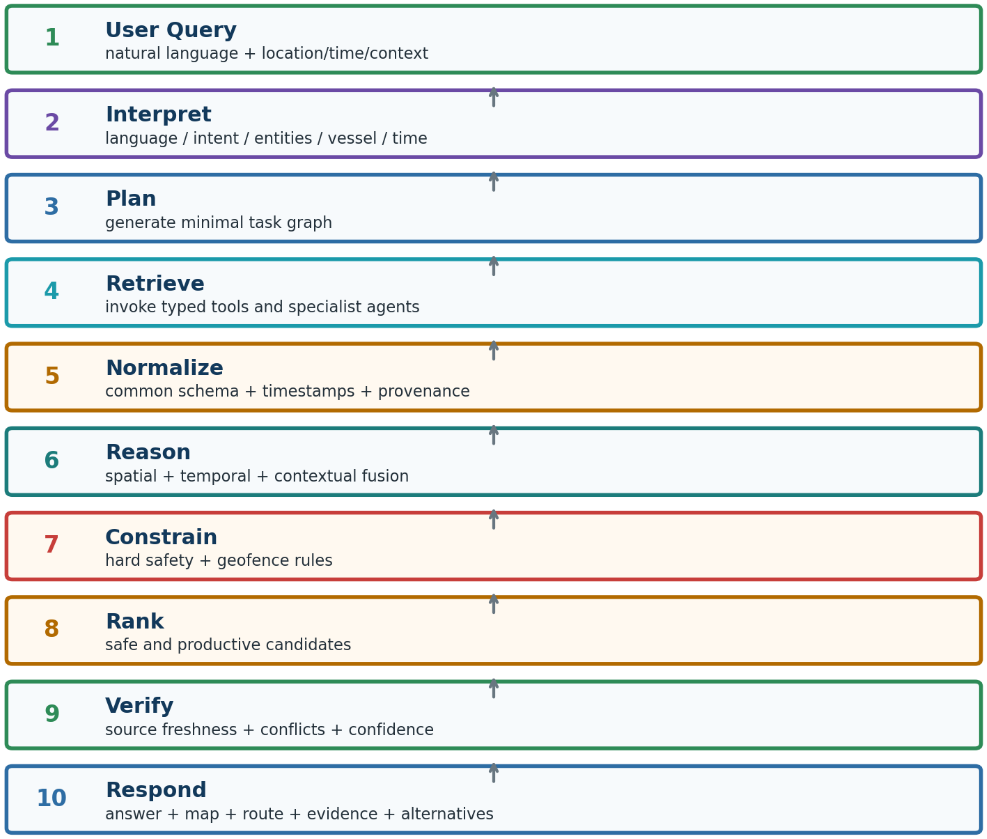
ORCA (Operational Reasoning & Collaborative Agents for Marine Intelligence) is our proposed solution. We are not trying to replace INCOIS, MOSDAC or IMD. We are building an intelligent reasoning layer above them.

Simple example

A fisherman asks: "I have a 5 m boat. Can I leave Malpe at 5 AM tomorrow and fish within 30 km?"

1. ORCA identifies the location, time, activity and vessel information.
2. The planner decides which information is required.
3. Marine, weather, hazard and geospatial agents retrieve the required data.
4. The system removes unsafe or restricted candidate areas first.
5. It ranks the remaining options using fishing and safety factors.
6. It returns a recommendation, route/map, reasons, source timestamps and alerts.

Our main differentiator: Safety and feasibility are checked first. Only after unsafe/restricted options are removed do we rank the remaining fishing opportunities.



Core principle: the LLM plans and explains; deterministic tools compute and enforce safety constraints.

End-to-end ORCA flow: question → planning → data/agents → reasoning → recommendation.

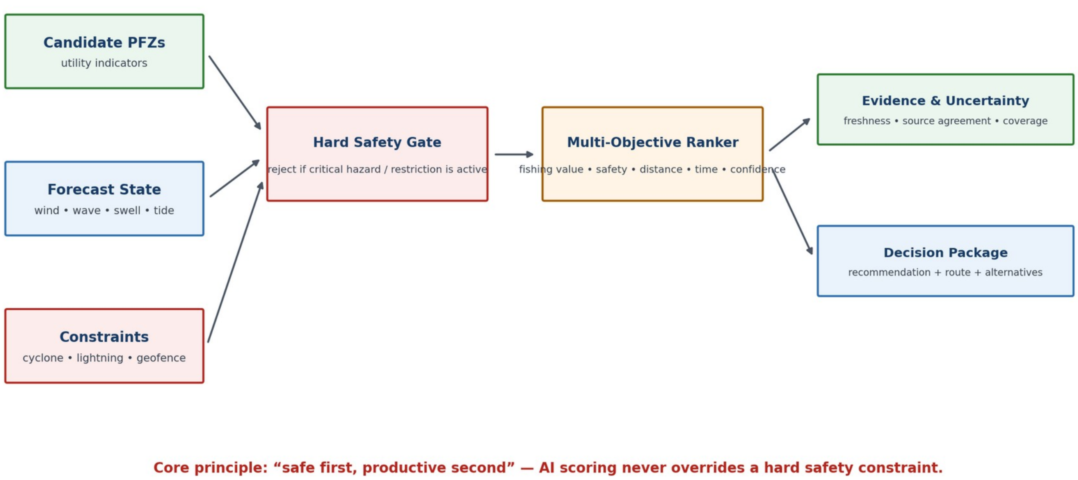
3. Why our solution should stand out

This problem statement is likely to attract many teams, so a simple “chatbot + map + weather + PFZ” system will look similar to other submissions. Our differentiation is in how the system makes the decision, not just how it displays data.

Common approach	ORCA approach
One chatbot calling a few APIs	Planner creates a task flow and calls only the required tools
Separate data values	Combines ocean + weather + hazards + GIS in one decision
AI gives a recommendation	Hard safety/geofence rules protect the final recommendation
Map shows layers	Map shows recommended zone, avoided areas and route
Answer is hard to verify	Every important result carries source and timestamp
Large feature list	Focused MVP around a few high-value marine decisions

Our unique “decision loop”

- Understand the user context → location, time, activity, vessel and constraints.
- Plan only the necessary tasks → avoid calling every dataset every time.
- Check hard constraints → hazards, restricted areas, critical missing data.
- Rank feasible options → fishing value, route quality and user requirements.
- Explain the decision → evidence, confidence, freshness and reasons.



Safe-first decision logic used by ORCA.

4. Data sources we can plan to use

The first build will prioritize official or authoritative sources. The exact API/end-point will be tested before final implementation because public availability and production access can differ.

Source	Data we need	Why we need it	Planned use
INCOIS	PFZ, ocean forecasts, tides, vessel/sea-state information	Operational marine intelligence	Fishing zones, safety, route planning
MOSDAC / ISRO	SST, ocean-colour/chlorophyll and satellite products	Earth Observation	Ocean analytics and zone ranking
IMD	weather, warnings, lightning/cyclone information	Meteorological safety	Hazard alerts and departure checks
INCOIS ERDDAP	machine-readable ocean datasets	Programmatic retrieval	Agent tools and historical/context analysis
GIS / Government boundaries	maritime limits, restricted/protected areas	Geofencing	Route/zone exclusion

Important engineering decision: Agents should not scrape websites individually. A common Marine Data Gateway will normalize data from different sources into one internal format.

Example internal data record

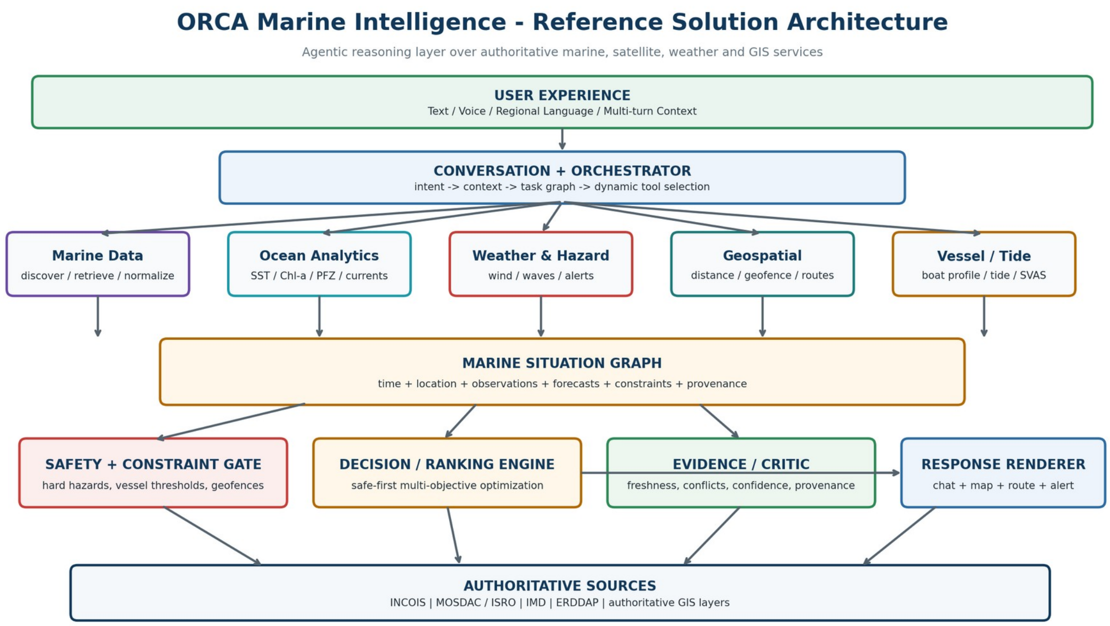
```
Parameter | value | unit | latitude | longitude | valid time | source | observed/forecast |
quality
```

Why this is feasible

- The project can begin with a Karnataka-coast pilot instead of the whole Indian coastline.
- We can use official operational products instead of building new weather/ocean models.
- The data gateway hides different formats and APIs from the agents.
- Historical data can be added later for analytics and ML.

5. Solution architecture

The architecture has four simple parts: user interface, agent layer, data/tools layer, and decision layer.



ORCA solution architecture.

Main components

Component	Role
UI / Dashboard	Chat, map, alerts, route, evidence and status.
Planner Agent	Understands the request and creates the task flow.
Marine Data Agent	Finds PFZ/ocean/satellite information through tools.
Weather & Hazard Agent	Checks forecast, warnings, lightning, cyclone and wave risk.
Geospatial Agent	Handles nearest/within/intersection/route/geofence calculations.
Risk & Decision Agent	Applies safety rules and ranks feasible options.
Evidence Agent	Checks data freshness, source provenance and consistency.
Marine Data Gateway	Common interface to external APIs/data services.
GIS / Database	PostgreSQL/PostGIS for spatial data and cached information.

6. How we will build the agents

We will not make agents “chat with each other” freely. Each agent will be a small software component with a clear input, tools and output. This makes the system easier to test.

Agent responsibilities

Agent	What it does	Example task
Planner	Breaks a user request into tasks	“Safety + fishing + route”
Marine Data	Retrieves marine/satellite data	Get PFZ, SST, chlorophyll
Weather/Hazard	Retrieves weather and hazard status	Check wind, wave, cyclone
Geospatial	Performs spatial calculations	Nearest PFZ, route intersection
Risk/Decision	Applies rules and ranking	Reject unsafe zones
Evidence	Checks provenance and freshness	Show source/time/conflict
Conversation	Maintains language and context	Kannada follow-up query

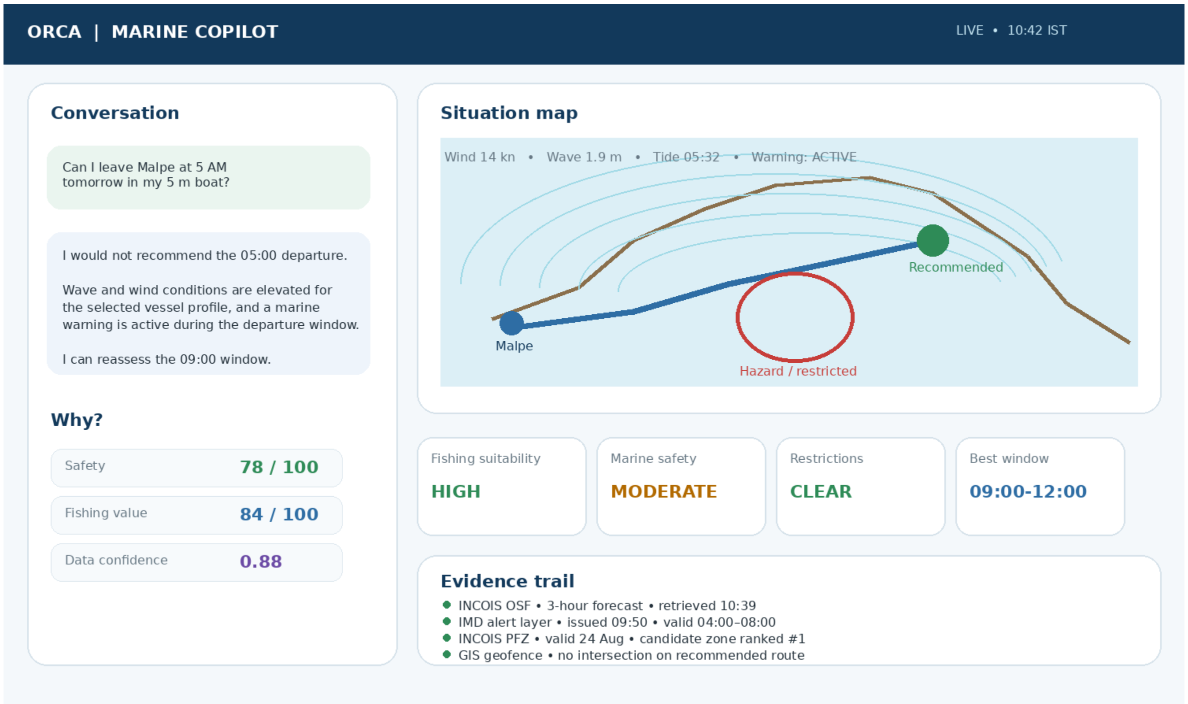
How the reasoning works

7. Parse the question: who, where, when, what activity and what limits?
8. Create a task plan.
9. Run independent data tasks in parallel where possible.
10. Normalize results into a common marine representation.
11. Apply hard safety/geofence checks.
12. Rank the remaining options.
13. Verify the recommendation against the evidence.
14. Generate the final answer and map.

Important: The LLM decides what to do and how to explain it. Deterministic GIS calculations, safety rules and data validation decide what is allowed.

7. UI/UX and dashboard

The interface should look like a simple marine operations dashboard, not a generic AI chat screen. The user should always see the recommendation, map and reason together.



Illustrative ORCA dashboard concept.

Main dashboard areas

- Chat area — user asks questions naturally.
- Understanding chips — location, time, vessel and intent detected by the system.
- Recommendation card — “Go / Avoid / Reassess / Best zone”.
- Map — route, PFZ, hazards and restricted areas.
- Why this answer? — short evidence summary and source timestamps.
- Alerts — active cyclone, lightning, high wave or geofence warning.
- Scenario controls — change time/location/vessel and recompute.

Multilingual approach

For the first prototype we propose English + Kannada. The language detector identifies the query language, the internal reasoning remains structured, and the final response is generated in the same language.

8. Technical approach

Area	Planned technologies
Backend	Python, FastAPI
Agent orchestration	LangGraph or equivalent graph/workflow orchestrator
LLM	API-based model for planning, tool use and language
Database	PostgreSQL + PostGIS
Data processing	Pandas, NumPy, GeoPandas, Shapely
ML	Risk/score models where useful; transparent baseline first
Computer Vision	Optional satellite-image analysis for later phases
Frontend	React / Next.js
Maps	MapLibre / Leaflet / compatible map stack
Deployment	Docker + cloud VM/container; local demo fallback

9. Feasibility

- Start with one geography: Karnataka coast.
- Start with 4–5 user journeys instead of every PS feature.
- Use existing operational datasets instead of predicting weather or ocean conditions from scratch.
- Build one stable data gateway before adding many agents.
- Use rule-based safety first; add ML only where it improves ranking or anomaly detection.
- Keep the first demo focused on real data, clear evidence and a working end-to-end flow.

MVP user journeys

15. Can I safely go fishing tomorrow morning?
16. Where is the nearest useful PFZ?
17. What are the sea conditions near me?
18. Find a productive area that is also safe and not restricted.
19. Show the safest route to the selected area.

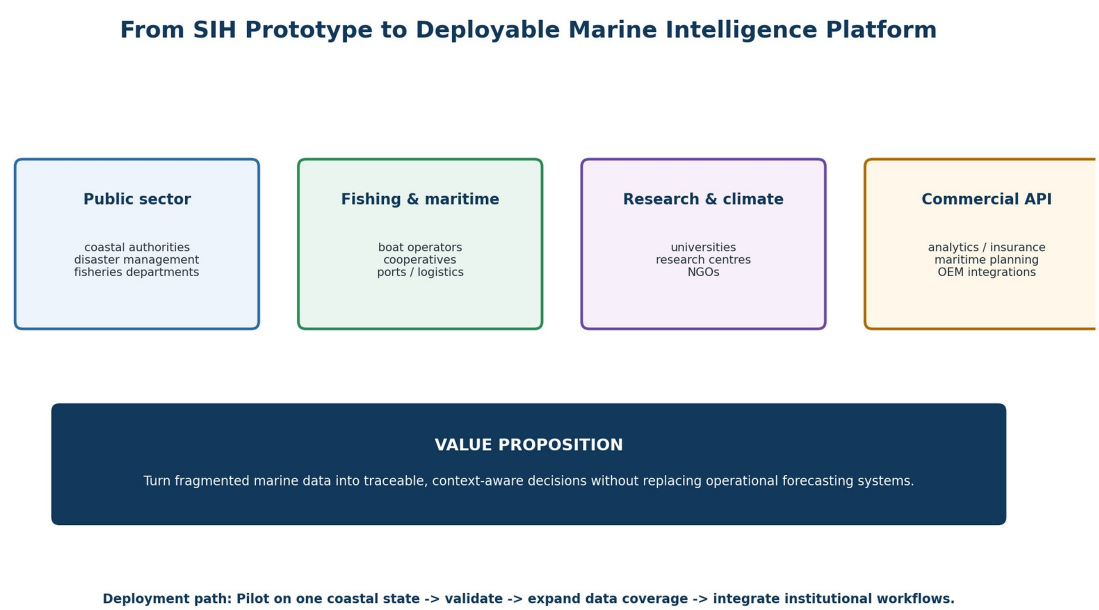
10. Viability and real-world use

The same platform can serve multiple stakeholders because the underlying data and reasoning layer are shared. What changes is the user role and the dashboard view.

User	Need	ORCA value
Fishermen	Safe and productive trips	Natural-language advice, PFZ, alerts, route and geofencing
Coastal authorities	Situation awareness	Hazard map, affected areas, summaries and alerts
Researchers	Marine analysis	Cross-source queries and historical comparison
Disaster teams	Fast marine risk picture	Warnings, affected zones and spatial summaries
Maritime operators	Operational planning	Route risk and marine conditions

Possible deployment model

- Pilot with a coastal institution or research partner.
- Provide role-based dashboards for different stakeholders.
- Expose decision APIs so existing mobile or maritime systems can consume ORCA outputs.
- Offer notification workflows for alerts and geofence events.
- Scale from a regional pilot to wider coastal coverage as data access and validation improve.



Illustrative stakeholder/deployment model.

11. Implementation roadmap

Phase	What we build	Output
1. Data foundation	Test INCOIS/MOSDAC/IMD access; normalize core datasets	Working Marine Data Gateway
2. Core agents	Planner + marine + weather/hazard + geospatial + risk	First end-to-end query works
3. Decision engine	Safety gates + ranking + provenance	Explainable recommendation
4. Dashboard	Chat + map + alerts + evidence	Demo-ready product
5. Validation	Test query benchmark; fix failures; add Kannada	Reliable SIH demo
6. Extension	Satellite CV, richer routing, more regions	Post-MVP roadmap

What would make the mentor confident?

- A working real-data demo instead of a mock chatbot.
- A small architecture that can actually be implemented by the team.
- Clear separation between AI reasoning and deterministic safety/GIS logic.
- Evidence and timestamps for every recommendation.
- A focused MVP that can be expanded after the core loop is stable.

Recommended mentor decision: Approve a Karnataka-coast pilot, a safety-first architecture, official data sources as the foundation, and a 5-scenario MVP. Then the team can start building the data gateway and planner.

12. Research references

- **INCOIS – Potential Fishing Zone Advisory —** https://www.incois.gov.in/MarineFisheries/PfzAdvisory
- **INCOIS – PFZ WebGIS —** https://www.incois.gov.in/MarineFisheries/PfzWebGis
- **INCOIS – Ocean State Forecast —** https://www.incois.gov.in/site/services/osf.jsp
- **INCOIS – Small Vessel Advisory —** https://www.incois.gov.in/site/services/SVA_overview.jsp
- **INCOIS ERDDAP —** https://erddap.incois.gov.in/erddap/index.html
- **MOSDAC – Oceansat-3 / EOS-06 —** https://www.mosdac.gov.in/oceansat-3
- **MOSDAC – Data Download API —** https://mosdac.gov.in/downloadapi-manual
- **ISRO – EOS-06 overview —** https://www.isro.gov.in/EOS_06.html
- **IMD – APIs / services —** https://mausam.imd.gov.in/responsive/apis.php
- **OceanAI (2025) —** https://arxiv.org/abs/2511.01019
- **Sensorium Arc (2025) —** https://arxiv.org/abs/2511.15997

Note: data access, API availability, authentication, update frequency and usage permissions will be validated during the first engineering sprint before the final implementation is frozen.